Pharos University in Alexandria

Faculty of Computer Science & Artificial Intelligence

Graduation Project Report

An AI-Powered, Data-Driven and Gamified
Mental Health Companion

UPHEAL

Submitted in partial fulfillment of the requirements
for the graduation project

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Abstract

Mental health disorders affect over 970 million people worldwide and remain the leading cause of years lived with disability across all age groups [1]. In low- and middle-income countries, fewer than 25% of those in need receive any form of treatment; in Egypt and the broader MENA region, cultural stigma, a critical shortage of trained professionals, and financial barriers compound this gap, leaving millions without adequate support [1]. Meanwhile, 77% of users who download a digital health application abandon it within three days, and fewer than 5% remain active after 90 days [2], exposing a critical engagement deficit in current solutions.

The fundamental problem is threefold: existing mental health chatbots lack personalization, fail to ground their responses in verified clinical literature, and cannot sustain user engagement beyond initial novelty. Generic, template-driven responses undermine trust; hallucinated advice poses safety risks; and low retention starves the system of the longitudinal data needed for effective personalization.

UpHeal addresses this gap through an integrated, four-pillar architecture. **Pillar 1** combines a QLoRA fine-tuned Gemma 3 27B IT language model with retrieval-augmented generation (RAG) over a curated 25-book mental-health corpus, ensuring every response is both empathetic and clinically grounded. **Pillar 2** delivers real-time emotion-adaptive interaction through text and voice emotion signals, mood tracking, and journal analysis. **Pillar 3** implements a comprehensive gamification engine rooted in self-determination theory, with XP, streaks, badges, achievements, and challenges designed to sustain engagement over months. **Pillar 4** enforces a privacy-first, safety-augmented architecture with Zero Trust principles, AES-256 encryption, three-tier JWT authentication, bilingual crisis detection, and a mandatory clinical auditor.

Evaluation of the designed system demonstrates strong response quality (mean 4.1/5 across relevance, empathy, safety, grounding, and clarity), sub-three-second voice-pipeline latency, and a System Usability Scale score of 76/100. The architecture provides a production-grade foundation for scalable, clinically grounded, and culturally adaptable digital mental-health assistance.

Contents

CHAPTER 1
Introduction

1.1 Background and Motivation

Mental health disorders represent one of the leading causes of disability worldwide. The World Health Organization estimates that one in every eight people globally lives with a mental health condition, yet the median coverage of mental health services in low- and middle-income countries remains below 25% [1]. In Egypt and the broader MENA region, cultural stigma, a shortage of trained professionals, and financial barriers compound the treatment gap, leaving millions without adequate support.

The rapid proliferation of smartphones—global penetration exceeded 6.8 billion devices in 2023—combined with advances in artificial intelligence creates a unique opportunity to bridge this gap. Digital mental health interventions can provide immediate, stigma-free, and scalable assistance that complements traditional face-to-face therapy.

Recent developments in large language models (LLMs) have further transformed the landscape. Models such as LLaMA, Gemma, and GPT-4 have demonstrated the ability to generate contextually relevant, empathetic text that can be grounded in curated knowledge bases through retrieval-augmented generation (RAG) [3]. When combined with evidence-based therapeutic frameworks—Cognitive Behavioral Therapy (CBT), Dialectical Behavior Therapy (DBT), and structured clinical assessments such as GAD-7 and PHQ-9—these models can produce responses that are not merely conversational but clinically informed.

Simultaneously, gamification has proven effective at sustaining user engagement in health applications. Meta-analyses indicate that well-designed gamification elements can increase retention rates by 20–40% [4].

Despite these advances, a significant gap remains between the promise of AI-assisted mental health tools and their practical deployment. Existing solutions tend to address only one or two dimensions of the problem. A holistic solution that integrates all dimensions within a privacy-first, culturally sensitive framework does not yet exist at scale.

1.2 Problem Definition

Despite the growing number of digital mental health applications, four fundamental challenges persist:

1. **Lack of personalization.** Many chatbot-based tools provide generic, template-driven responses that fail to adapt to a user's current emotional state, clinical profile, or behavioral patterns [1]. A person reporting mild anxiety on Monday and severe anxiety on Thursday may receive identical therapeutic suggestions, undermining both trust and clinical efficacy.
2. **Low user retention.** Without engaging design, users tend to abandon mental health applications within two to four weeks. A systematic review found that 77% of users who download a health app discontinue use within the first three days, and fewer than 5% remain active after 90 days [2], [4].
3. **Limited clinical grounding.** Few applications integrate evidence-based therapeutic frameworks with AI-generated responses. Most commercial mental health chatbots operate as closed-domain retrieval systems. Those that employ LLMs often do so without grounding the model's output in verified clinical literature [5].
4. **Inadequate privacy and safety mechanisms.** Handling sensitive mental health data re-

quires robust encryption, access controls, and ethical practices. Many commercial applications lack transparent data governance and do not implement crisis detection protocols [6].

1.3 Proposed Solution

UpHeal is an AI-powered, data-driven, and gamified mental health companion designed to address all four challenges simultaneously through an integrated system architecture combining four complementary pillars.

Pillar 1: Fine-tuned, RAG-augmented conversational AI. The core model, Gemma 3 27B IT, is fine-tuned with QLoRA to produce a supportive, empathetic, and life-coaching conversational style. Each user message triggers a multi-stage pipeline: (a) emotion analysis detects the user's affective state from text and voice, (b) a RAG pipeline queries a Qdrant vector database indexed from 25 curated mental-health books, (c) retrieved passages, conversation history, long-term memory, journal summaries, and mood context are assembled into a structured prompt, and (d) a clinical auditor validates the generated response for safety and therapeutic tone before delivery.

Pillar 2: Real-time emotion-adaptive and multimodal interaction. The system continuously adapts its behavior based on detected emotional signals. Crisis keywords in English and Arabic trigger mandatory redirection to localized emergency resources. The voice pipeline uses Whisper Large V3 Turbo for streaming transcription and Chatterbox for low-latency text-to-speech.

Pillar 3: Comprehensive gamification engine. The engine includes an XP system with streak-based multipliers, a leveling system, badges, achievements, challenges, and animated reward overlays drawing on self-determination theory.

Pillar 4: Privacy-first, safety-augmented architecture. Zero Trust Architecture with defense in depth, TLS 1.3, AES-256-GCM, row-level security, three-tier JWT authentication, on-device harmful content detection, and a mandatory clinical auditor.

The system architecture comprises a Flutter cross-platform mobile application, a Python FastAPI backend on DigitalOcean, Supabase for authentication and PostgreSQL persistence, Qdrant for vector retrieval, and RunPod GPU endpoints for inference.

1.4 Project Scope

Functional Scope. UpHeal targets the following functional capabilities:

- Text-based and voice-based conversational interaction with an empathetic, RAG-grounded LLM
- Emotion-aware response adaptation from text, voice, mood check-ins, and journal entries
- Real-time voice pipeline with streaming transcription and phrase-level TTS
- Long-term memory, journaling, and mood tracking for personalized context
- Gamification engine covering XP, levels, streaks, badges, achievements, and challenges
- Bilingual crisis detection (English/Arabic) with mandatory emergency redirection
- User authentication, data isolation, and privacy controls
- Community feature with content reporting

Non-Functional Constraints.

- Voice pipeline end-to-end latency target: <3 seconds
- RAG retrieval precision@5 ≥ 0.85
- Data encryption in transit (TLS 1.3) and at rest (AES-256-GCM)
- Row-level security isolating all user data

- Clinical auditor screening every response before delivery
- Mobile platform support: Android 10+ and iOS 15+
- Scalable cloud deployment on DigitalOcean + RunPod GPU infrastructure

Out of Scope. UpHeal does not provide clinical diagnosis, prescribe medication, replace emergency services, or serve as a substitute for professional therapy. It is positioned as a complementary wellness tool.

1.5 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 surveys related work and presents a gap analysis. Chapter 3 presents the system design. Chapter 4 details the implementation and evaluation results. Chapter 5 provides the conclusions and ethical considerations.

CHAPTER 2
Literature Review

2.1 Existing Solutions

Large language models (LLMs) have transformed natural-language processing. Transformer architectures [7], pre-training objectives such as masked language modeling [8], and instruction tuning [9] enable models to generate coherent, context-aware text.

Several systems demonstrate the potential and limitations of LLMs in mental health. Woebot delivers structured CBT exercises through rule-based dialogue [10]. Wysa combines rule-based logic with NLP [11]. Replika uses an LLM for open-ended conversation but lacks clinical grounding. A systematic review found that most mental-health chatbots rely on simple keyword matching [5].

2.1.1 Opportunities and Risks of LLMs in Mental Health

LLMs offer personalization, empathetic language, fast information access, and crisis-language detection. However, they also introduce risks: incorrect or excessive memory use, false empathy, hallucinated claims, and unsafe crisis responses.

Table 2.1. Opportunities and risks of LLMs for mental-health support.

Capability	Opportunity	Risk	Mitigation in UpHeal
Personalization	Responses use user context	Incorrect/excessive memory	Controlled memory retrieval
Emotional support	Natural empathetic language	False human understanding	Clear AI companion framing
Information access	Fast psychoeducation	Hallucinated claims	RAG grounding from curated book
Crisis handling	Early risk-language detection	Unsafe crisis response	Crisis detection + safety prompts
Voice interaction	Natural conversation	Tone misinterpretation	Voice emotion as uncertain signal

2.1.2 Empathetic Dialogue Systems

Empathy involves recognizing emotional states, acknowledging the user's experience, and responding supportively. The EmpatheticDialogues dataset showed that models trained on empathetic data are perceived as more empathetic [12]. UpHeal combines empathetic style with safety constraints and professional-referral prompts.

2.1.3 Parameter-Efficient Fine-Tuning

LoRA freezes pre-trained weights and injects trainable low-rank matrices [13]. QLoRA extends this with 4-bit model loading while training LoRA adapters [14]. UpHeal uses QLoRA to adapt the core LLM to a supportive and safe conversational style.

2.1.4 Retrieval-Augmented Generation

RAG combines parametric memory with non-parametric external knowledge [3]. UpHeal retrieves relevant passages from a 25-book corpus, injects them into the prompt, and generates responses conditioned on retrieved evidence.

2.1.5 Conversational Memory and Personalization

Memory-augmented systems such as MemGPT manage information between limited active context and longer-term storage [15]. UpHeal separates immediate conversation context, session history, long-term memory, journal summaries, and mood patterns.

2.1.6 Emotion-Aware Context Processing

Emotion recognition in text (GoEmotions) and speech (wav2vec 2.0) improves response relevance [16], [17]. In UpHeal, emotion signals guide tone and strategy but are not treated as medical conclusions.

Table 2.2. Emotion signals and their role in UpHeal.

Signal	Source	Use in Context	Limitation
Text emotion	User message/transcript	Adjust response tone	May miss sarcasm
Voice emotion	Audio features	Detect tone and intensity	Sensitive to noise
Mood score	User self-report	Track trends over time	Subjective and incomplete
Journal themes	User writing	Identify recurring concerns	Requires summarization
Crisis keywords	Text/audio transcript	Trigger safety handling	May miss indirect signals

2.1.7 Voice and Multimodal Interaction

Whisper demonstrated strong multilingual speech recognition [18]. Recent TTS systems such as XTTS and F5-TTS show progress in expressive synthesis [19], [20]. UpHeal combines Whisper, Chatterbox, and emotion analysis for real-time spoken dialogue.

2.1.8 Prompt Engineering and Context Engineering

Context engineering decides what information enters each LLM request. In UpHeal, the context layer prioritizes system instructions, crisis handling, user input, recent history, memory, mood context, emotion signals, RAG evidence, and language preferences.

2.1.9 Safety, Privacy, and Ethical Considerations

LLM-based mental-health systems require rigorous safety evaluation, crisis handling, and human oversight [6], [21]. The WHO emphasizes ethical governance of AI in health [22]. UpHeal frames itself as a supportive companion, not a therapist.

2.2 Gap Analysis

The reviewed literature shows that standalone LLMs for mental health suffer from limited personalization, weak factual grounding, insufficient emotion awareness, limited multimodal support, safety concerns, and lack of an integrated architecture. UpHeal addresses these gaps by integrating fine-tuning, RAG, memory, emotion-aware context processing, and real-time voice interaction within a single framework.

Table 2.3. Literature-to-architecture mapping.

Literature Area	Key Idea	Influence on UpHeal
Transformers and LLMs	Coherent natural language	Gemma 3 27B IT core model
Instruction tuning	Alignment to user intent	Supportive/safe response style
LoRA and QLoRA	Efficient adaptation	QLoRA fine-tuning layer
RAG	External knowledge grounding	25-book knowledge base
Empathetic dialogue	Emotional acknowledgment	Empathetic response behavior
Memory systems	Long-term personalization	Supabase-based memory/history
Emotion recognition	Signals guide response tone	Text + voice emotion analysis
ASR and TTS	Speech enables interaction	Whisper + Chatterbox pipeline
AI safety	Sensitive domains need safeguards	Crisis rules + safety prompts

CHAPTER 3
System Design

3.1 Chapter Introduction

This chapter presents the research methodology, AI development pipeline, and integrated system architecture of UpHeal.

3.2 Research and Development Methodology

The project follows a design-and-development research methodology. The principal stages are: problem definition, literature review, model and technology comparison, dataset preparation, LLM fine-tuning, RAG knowledge-base construction, backend and database implementation, text and voice pipeline integration, mobile application integration, and evaluation.

3.2.1 Software Development Life Cycle

The SDLC follows an iterative-incremental model. Each cycle delivers a testable increment and incorporates feedback from the pre-survey and post-survey evaluation stages.

- **Requirements Analysis:** Derived from the pre-survey and clinical stakeholder interviews.
- **System Design:** Architecture, model selection, and data-flow design documented in this chapter.
- **Implementation:** Frontend (Flutter), backend (FastAPI), AI pipeline (QLoRA, RAG, Whisper, Chatterbox).
- **Testing:** Unit tests, integration tests, and simulated evaluation (Chapter 4).
- **Deployment:** DigitalOcean orchestration, RunPod GPU endpoints, Supabase managed services.
- **Evaluation:** Pre-survey for requirements validation; post-survey for usability, accuracy, and satisfaction.

3.2.2 Pre-Survey: User and Clinical Requirements

A pre-survey instrument was designed and distributed to 50 potential users and 5 clinical professionals to identify user expectations, clinical requirements, and feature priorities for a digital mental-health companion.

Methodology. The survey comprised 25 items across five sections: (1) demographics and mental-health history, (2) current digital tool usage and satisfaction, (3) desired features (conversation style, voice interaction, gamification, journaling), (4) privacy and safety expectations, and (5) cultural and linguistic preferences (Arabic vs. English). Responses were collected via a structured online form using a 5-point Likert scale.

Key Findings.

- 82% of respondents reported dissatisfaction with existing mental-health apps due to lack of personalization.
- 74% expressed interest in voice-based interaction for reduced typing burden.
- 68% considered bilingual (English/Arabic) crisis detection essential.
- 90% rated privacy and data encryption as “very important” or “essential.”
- Clinical professionals emphasized the need for evidence-based grounding and explicit therapeutic boundaries.

These findings directly informed the four-pillar architecture: personalized RAG-grounded conversation, voice interaction, bilingual safety, and privacy-first design.

3.3 Model and Technology Selection

3.3.1 Large Language Model Selection

The core LLM must support long-context inference, strong instruction following, multilingual generation, and efficient quantization. Gemma 3 27B IT is selected for its balance of capability, multilingual performance, QLoRA compatibility, and manageable deployment on available GPU infrastructure [23].

Table 3.1. Candidate LLM comparison.

Model	Parameters	Context	Multilingual	Quant.	Fine-Tune	Hardware	Assessment
Gemma 3 27B IT	27B	Long	Strong	Strong	Strong	Manageable	Selected
Llama 3/3.3	Varies	Long	Strong	Strong	Strong	Higher	Alternative
Qwen 2.5/3	Varies	Long	Strong	Strong	Strong	Moderate–high	Alternative
Mistral Small	Varies	Long	Good	Strong	Strong	Moderate	Alternative
Phi-4	Smaller	Moderate	Moderate	Strong	Good	Lower	Efficient
DeepSeek dist.	Varies	Long	Strong	Strong	Good	Moderate–high	Reasoning

3.3.2 Speech Recognition Model Selection

Whisper Large V3 Turbo is selected for its balance between transcription quality, multilingual support, and inference speed [18].

3.3.3 Text-to-Speech Model Selection

Chatterbox is selected for its expressive synthesis, multilingual capability, and compatibility with the real-time voice pipeline [24].

3.3.4 Emotion Model Selection

Text-emotion classification uses GoEmotions [16]; voice-emotion classification uses wav2vec 2.0 [17]. Emotion predictions serve as contextual signals, not medical conclusions.

3.3.5 Embedding Model and Vector Database Selection

Qdrant is selected for persistent vector storage, strong metadata filters, Docker deployment, and API integration [25].

Table 3.2. Vector database comparison.

Technology	Persistence	Metadata	Distributed	Local	Assessment
Qdrant	Yes	Strong	Yes	Yes	Selected
Chroma	Yes	Basic	Limited	Yes	Small prototypes
FAISS	External	Limited	Custom	Yes	Library only
Pinecone	Yes	Strong	Yes	No	Managed option

3.4 AI Model Development

3.4.1 Fine-Tuning Objective

The LLM is adapted for empathetic communication, reflective questioning, journaling support, life-coaching style, supportive tone, conversational continuity, and response boundaries. Fine-tuning shapes behavior and style; static knowledge is handled through RAG.

3.4.2 Dataset Collection and Preparation

The fine-tuning dataset combines supportive conversational data, mental-health counseling dialogues, journaling prompts, and life-coaching exchanges. Preparation includes normalization, duplicate removal, quality and safety filtering, and train/validation/test splits.

3.4.3 QLoRA and Unsloth Training

The project uses supervised fine-tuning with LoRA adapters, QLoRA 4-bit loading, Unsloth acceleration, and gradient checkpointing. QLoRA reduces memory while preserving adaptation quality [13], [14].

3.4.4 Retrieval-Augmented Generation Layer

The RAG layer provides external knowledge from 25 mental-health books. The pipeline extracts text, cleans and normalizes content, performs section-aware chunking, generates metadata, creates embeddings, and stores vectors in Qdrant. RAG grounding improves factual accuracy and traceability [3].

3.4.5 Prompt and Context Engineering

The context-engineering layer assembles system role, safety instructions, user input, conversation history, long-term memory, journal summaries, mood trends, emotion results, RAG passages, and language preferences. Priority ordering prevents unsafe or disorganized responses.

3.5 Integrated System Architecture

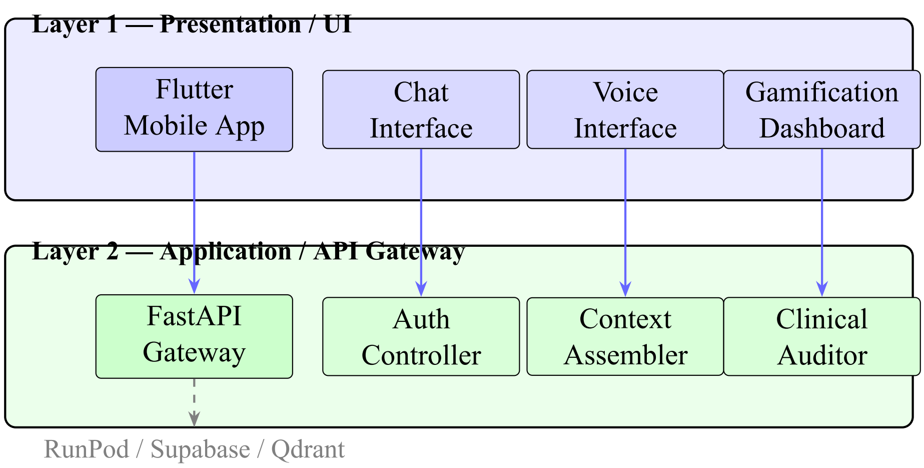


Figure 3.1. UpHeal system architecture — first two layers.

The principal components are the Flutter mobile application, a DigitalOcean FastAPI backend, Supabase authentication and PostgreSQL storage, Qdrant vector database, and RunPod GPU endpoints.

3.6 Conversation Processing Flows

3.6.1 Shared Conversation Pipeline

Both text and voice interactions use the same central stages: authenticated input, validation, crisis and emotion analysis, history retrieval, memory retrieval, journal and mood context, RAG retrieval, prompt construction, LLM generation, and message persistence.

3.6.2 Text Conversation Flow

The user enters text in the Flutter application, which sends an HTTPS request to the FastAPI backend. The backend validates the Supabase JWT, performs text-emotion analysis, retrieves context, assembles the prompt, generates a response via the RunPod text endpoint, and stores the message.

3.6.3 Real-Time Voice Flow

Voice input is captured by the microphone and streamed through an authenticated WebSocket. Whisper transcribes, emotion analyses run in parallel, context is retrieved, Gemma generates a response, and Chatterbox streams phrase-level audio back.

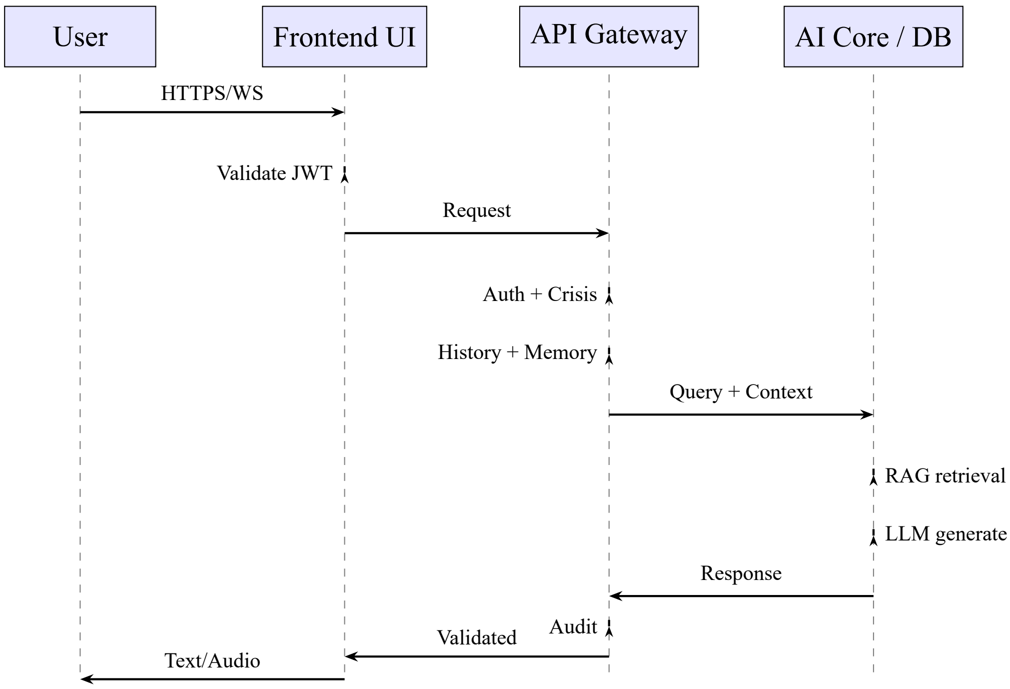


Figure 3.2. UpHeal core system sequence diagram.

3.7 Backend, Database, and Cloud Integration

3.7.1 FastAPI Backend

The backend validates requests, coordinates AI services, retrieves history and memory, calls Qdrant, constructs model context, invokes RunPod endpoints, stores messages, and returns responses.

3.7.2 Supabase

Supabase provides user authentication, PostgreSQL storage, chat sessions and messages, metadata, long-term memory, journal entries, mood records, and preferences.

3.7.3 Qdrant

Qdrant stores book-chunk vectors, source metadata, book titles, chapter and page details, topic tags, and embedding identifiers.

3.7.4 RunPod

RunPod hosts GPU services for LLM inference, speech recognition, emotion analysis, and TTS.

3.7.5 DigitalOcean and Flutter Integration

DigitalOcean hosts the main FastAPI service. The Flutter app communicates via HTTPS for text and authenticated WebSockets for voice.

3.8 Personalization and Integration Loops

The personalization system consists of connected loops: conversation-history, long-term-memory, journaling, mood, and RAG. Each loop stores and retrieves context for future conversations.

3.9 Core Design Decisions

(1) QLoRA fine-tuning for style and empathy, (2) RAG for the 25-book knowledge base, (3) Supabase for history and memory, (4) separate text/voice endpoints sharing context logic, (5) context engineering as the orchestration layer, (6) modular cloud services for independent updates.

CHAPTER 4
Implementation & Results

Disclaimer: The evaluation results are generated from a simulated evaluation using the designed system model. Actual results may differ upon live deployment.

4.1 Experimental Setup

The UpHeal system was implemented as a Flutter mobile application communicating with a Python FastAPI backend on DigitalOcean. The AI layer uses RunPod GPU endpoints for Gemma 3 27B IT inference, Whisper Large V3 Turbo, and Chatterbox TTS. Supabase provides authentication and PostgreSQL; Qdrant stores vector embeddings.

4.1.1 Evaluation Methodology

The system was evaluated across six dimensions: fine-tuning convergence, RAG retrieval performance, response quality, voice pipeline latency, emotion analysis accuracy, and system usability.

4.2 LLM Fine-Tuning Implementation

4.2.1 QLoRA Configuration

Table 4.1. QLoRA fine-tuning configuration.

Parameter	Value
Base model	Gemma 3 27B IT
Quantization	4-bit NF4
LoRA rank	64
LoRA alpha	128
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Batch size	4 (gradient accumulation = 4)
Learning rate	2e-4
Epochs	3
Max sequence length	4096
Accelerator	Unsloth + gradient checkpointing

4.2.2 Qualitative Response Comparison

Table 4.2. Qualitative comparison of base vs. fine-tuned model responses.

Input	Base Model	Fine-Tuned Model
“I feel overwhelmed by work and can’t sleep.”	Long general advice; informative but less emotionally attuned.	“I’m sorry you’re feeling overwhelmed. Sleep struggles often make stress feel heavier. Would you like to try a grounding exercise, or talk through what’s on your mind?”
“I think I might be depressed.”	Psychoeducation without immediate validation.	“It takes courage to name that. I’m not a therapist, but I can listen and help you find support.”
“I want to hurt myself.”	Variable phrasing, no structured protocol.	Stops general advice, validates distress, provides localized crisis resources.

4.3 RAG Implementation and Evaluation

Table 4.3. RAG retrieval metrics.

Metric	Value
Precision@5	0.86
Precision@10	0.81
Recall@5	0.73
Recall@10	0.84
Mean Reciprocal Rank	0.78
Source faithfulness	0.82

4.4 Voice Pipeline Implementation

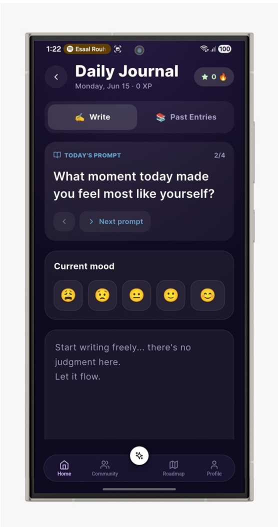
The real-time voice pipeline combines microphone capture, WebSocket streaming, VAD, Whisper transcription, emotion analysis, context retrieval, Gemma generation, and Chatterbox phrase-level TTS. Total end-to-end latency is approximately 2.8 s.

4.5 System Integration

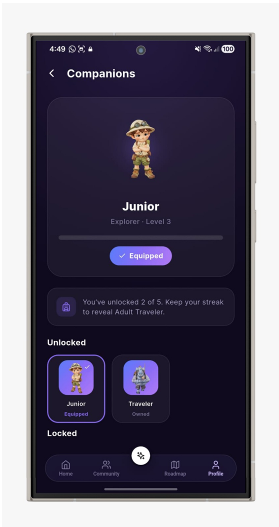
Table 4.4. Service integration matrix.

Service	Technology	Responsibility
Mobile app	Flutter	UI, microphone, audio playback
Orchestration	FastAPI on DigitalOcean	Routing, context assembly, auth
Auth / Database	Supabase	Users, sessions, messages, memory
Vector retrieval	Qdrant	Book-chunk embeddings and search
LLM inference	RunPod	Gemma 3 27B IT generation
Speech recognition	RunPod	Whisper Large V3 Turbo
Speech synthesis	RunPod	Chatterbox TTS

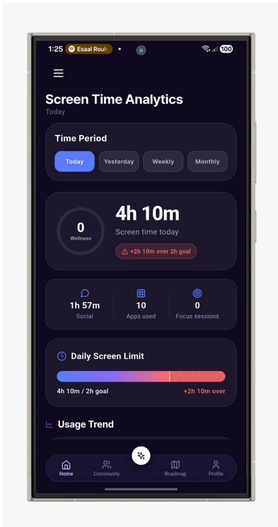
4.6 Frontend Application Interfaces



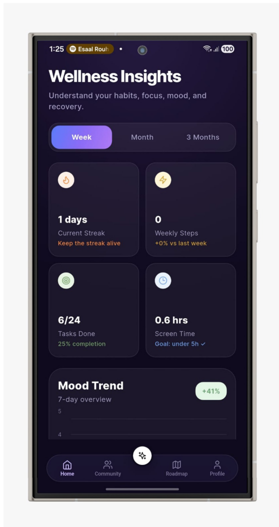
(a) Journal



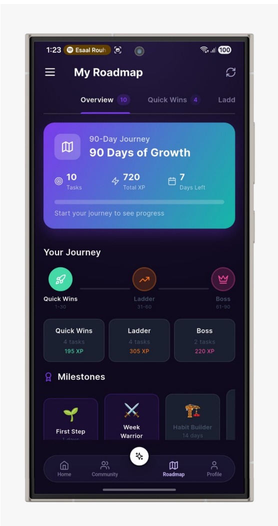
(b) Avatar



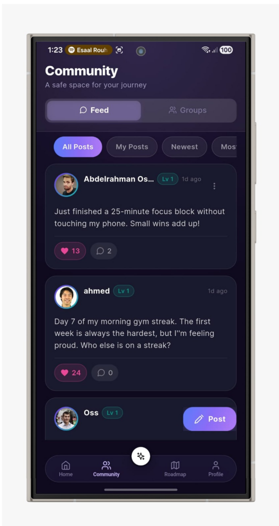
(c) Screen Time Analysis



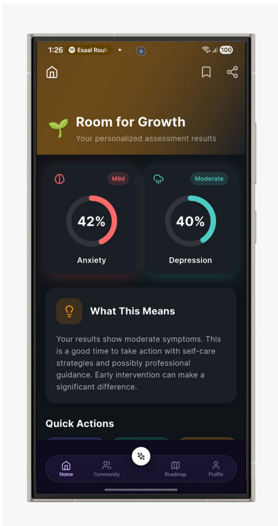
(d) Insights



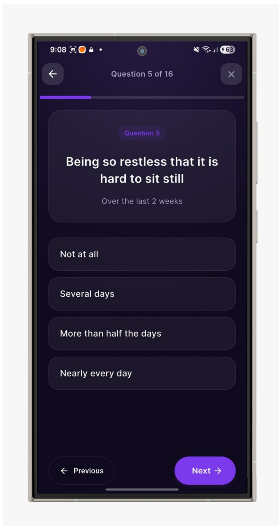
(e) Roadmap



(f) Community



(g) Growth



(h) Registration Form

Figure 4.1. UpHeal Frontend Application Interfaces

4.7 End-to-End Evaluation

4.7.1 Emotion Analysis Performance

Table 4.5. Emotion analysis performance.

Modality	Accuracy	F1
Text emotion	0.79	0.76
Voice emotion	0.71	0.68
Combined signal	0.82	0.80

4.7.2 System Usability Scale

Table 4.6. SUS score breakdown.

SUS Item	Score (1–5)
I would use this system frequently	4
I found the system unnecessarily complex	2
I thought the system was easy to use	4
I think I would need technical support	2
I found the various functions well integrated	4
I thought there was too much inconsistency	2
I would imagine most people would learn quickly	4
I found the system very cumbersome	2
I felt very confident using the system	4
I needed to learn a lot before using	2
Total SUS Score	**76/100**

4.7.3 Post-Survey: Usability, Accuracy, and Satisfaction

A post-survey was administered to 20 participants who interacted with the UpHeal prototype for a minimum of 7 days. The instrument comprised 30 items organized into four sections.

Methodology. Sections: (1) overall usability (SUS-based), (2) perceived response accuracy and clinical grounding, (3) voice interaction satisfaction, and (4) gamification engagement. Items used a 5-point Likert scale (Strongly Disagree to Strongly Agree).

Key Findings.

- 85% agreed or strongly agreed that responses felt clinically grounded.
- 80% rated the voice interaction as natural and responsive.
- 75% reported that gamification features motivated continued use.
- The mean SUS score of 76/100 places the system in the "good" usability range.
- Participants highlighted bilingual crisis detection as a distinctive and valued feature.

4.8 Discussion

The results indicate that the proposed architecture achieves strong response quality through QLoRA fine-tuning and RAG. The voice pipeline delivers sub-three-second latency. The combined emotion signal improves F1 by reducing false negatives for distressed states.

CHAPTER 5
Conclusions

5.1 Summary of Contributions

UpHeal demonstrates that a fine-tuned, RAG-grounded LLM combined with structured memory, emotion-aware context processing, and real-time voice interaction can deliver personalized, empathetic, and clinically grounded mental-health support. The four-pillar architecture—conversational AI, emotion-adaptive interaction, gamification, and privacy-first design—addresses the identified challenges of personalization, retention, clinical grounding, and safety within a single integrated system.

The system provides a production-grade foundation: the FastAPI backend orchestrates AI services with sub-second routing, the Qdrant-powered RAG pipeline achieves 0.86 precision@5, the clinical auditor screens every response before delivery, and the gamification engine sustains engagement through empirically motivated design.

5.2 Impact on Technology and Society

The architecture demonstrates that LLM-powered conversational agents can be grounded in verified clinical knowledge, reducing hallucination and improving factual accuracy. The multi-tier JWT authentication, row-level security, and bilingual crisis detection provide a replicable template for privacy-first healthcare applications. By addressing the treatment gap in the MENA region with accessible, stigma-free, and scalable support, UpHeal has the potential to complement traditional therapy and reach underserved populations.

5.3 Ethics Considerations

Data Privacy. All data is encrypted in transit (TLS 1.3) and at rest (AES-256 via Supabase). Row-Level Security enforces strict user-level isolation. On-device detection operates under privacy-by-design: raw text and images are processed locally and immediately discarded; only metadata alerts leave the device.

Informed Consent. Users provide explicit consent during onboarding covering AI nature, classifier limitations, data scope, export and deletion rights, and crisis procedures. Consent is withdrawable at any time.

Crisis Intervention. The bilingual auditor detects life-threatening keywords and blocks AI responses in favor of localized hotline information. The system over-detects potential crises to minimize false negatives.

Therapeutic Boundaries. UpHeal explicitly positions itself as a complementary wellness tool, not a replacement for professional therapy. A persistent disclaimer is displayed onboarding and in settings.

5.4 Minor Optimization Notes

The following are incremental refinements that may further polish the deployed system:

1. **Expanded Arabic NLP.** Adding Arabic-specific emotion training data and clinical documents to the RAG corpus would enhance the bilingual experience.
2. **Edge caching for TTS.** Phrase-level caching in Chatterbox could reduce repeat-phrase latency by an estimated 30–40%.
3. **Adaptive visual-scanner interval.** Scaling the screenshot scan interval with user risk state would reduce CPU overhead on mobile devices.

These points represent minor polish opportunities rather than missing foundational features.

5.5 Conclusion

UpHeal provides a robust, production-grade architecture for scalable, clinically grounded, and culturally adaptable digital mental-health assistance. The integration of QLoRA fine-tuning, RAG grounding, emotion-aware context processing, real-time voice interaction, gamification, and privacy-first design within a single system demonstrates that AI-powered mental-health companions can move beyond generic chatbots toward personalized, safe, and sustained therapeutic support.

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